

Talking with the Satellites

Building a Low-Cost Ground Station with SatNOGS

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today



satellite rise

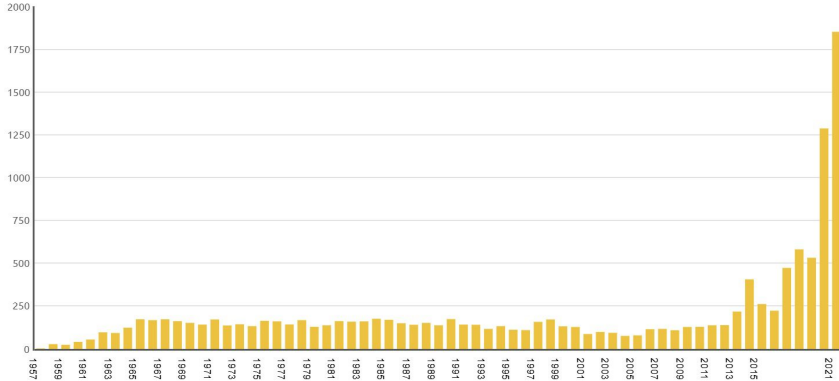
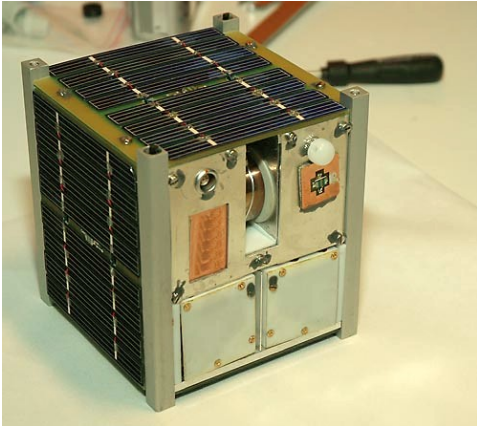


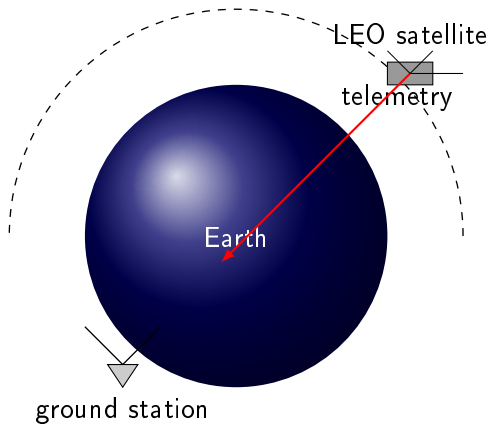
Figure: number of active satellites by year



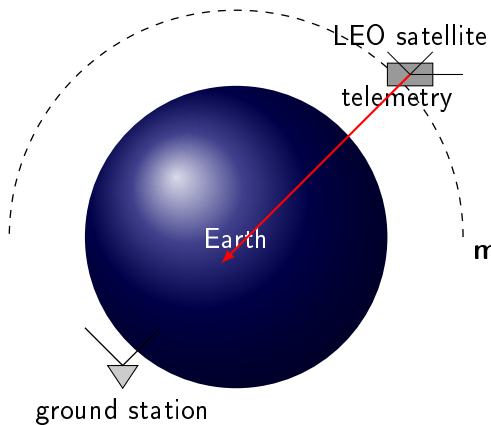
can i launch a satellite?



motivation



motivation

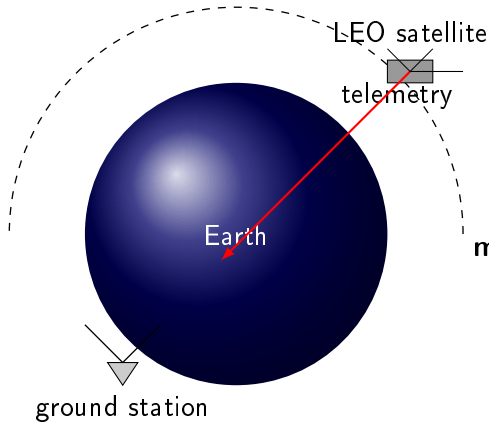


motivation:

- access real-time scientific data from space



motivation

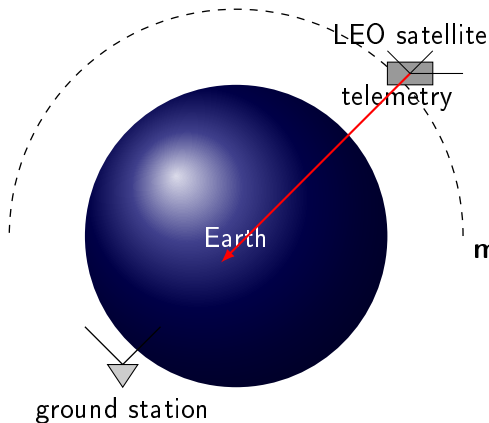


motivation:

- ▶ access real-time scientific data from space
- ▶ study RF propagation and satellite orbits



motivation

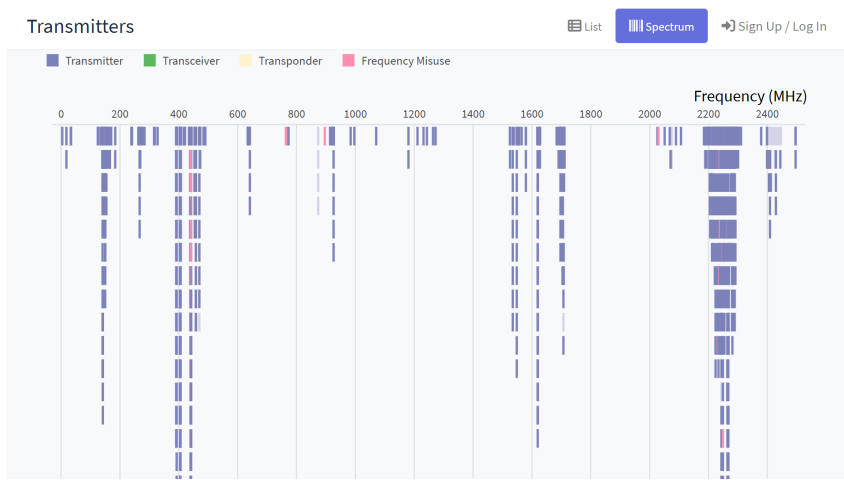


motivation:

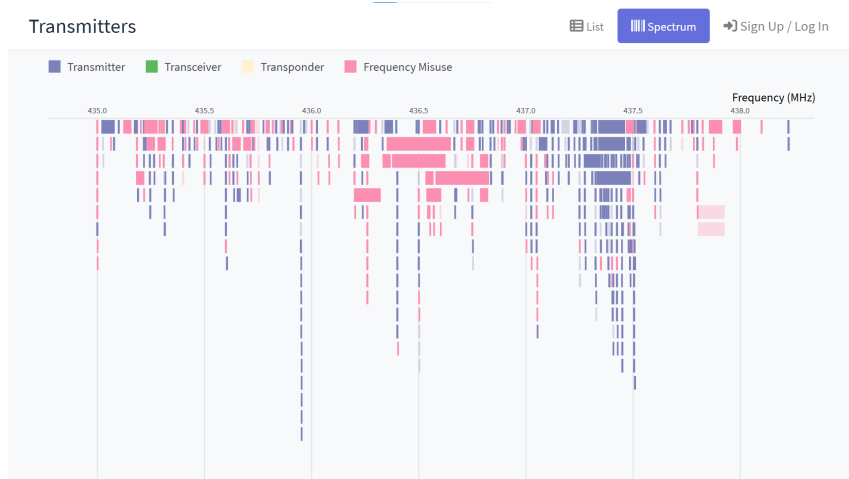
- ▶ access real-time scientific data from space
- ▶ study RF propagation and satellite orbits
- ▶ contribute to global open-source networks (SatNOGS)



listening to waves



amateur frequencies



hello
oooo●
hello

antenna
oooo

SatNOGS
oo

results
oooo

why tho?



setup

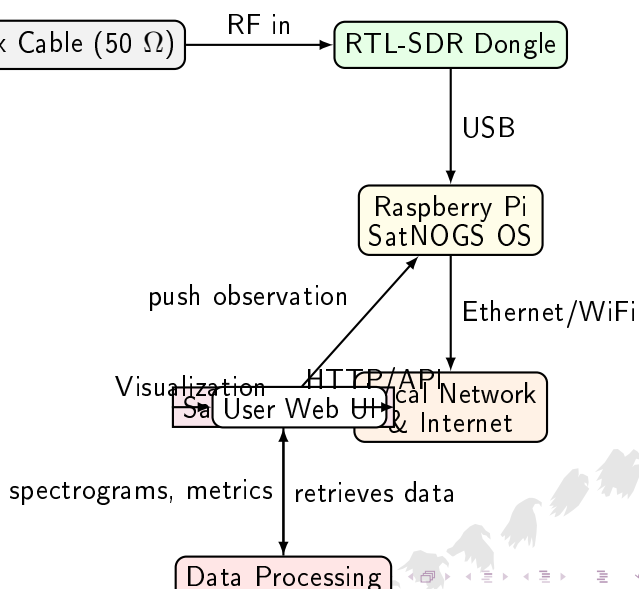


SatNOGS Station

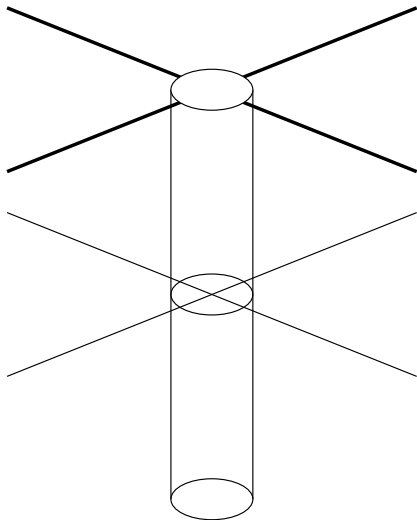
Ground Station RTL-SDR Dongle Antenna



setup



antenna



- ▶ active (upper) rod length:
 $L_a = \frac{\lambda}{4} = \frac{c}{4f} = 16.3 \text{ cm}$
- ▶ passive (lower) rod length:
 $L_p = \frac{\lambda}{2} = \frac{c}{2f} = 34.6 \text{ cm}$
- ▶ distance between rods:
 $d = \frac{3\lambda}{8} = \frac{3c}{8f} = 25.8 \text{ cm}$
- ▶ interference: constructive



antenna

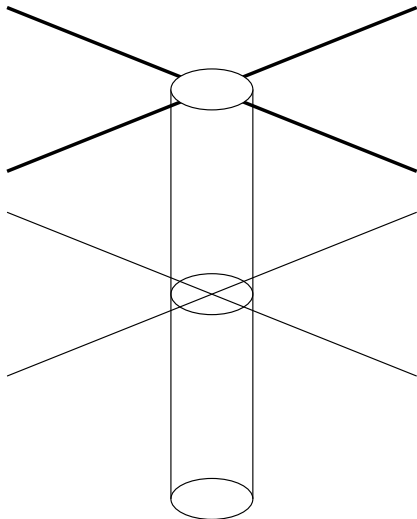


Figure: the antenna with RTL-SDR dongle just outside the Garage's window



sdr + computer



Figure: rtl-sdr dongle



Figure: raspberry pi



satnogs: concept

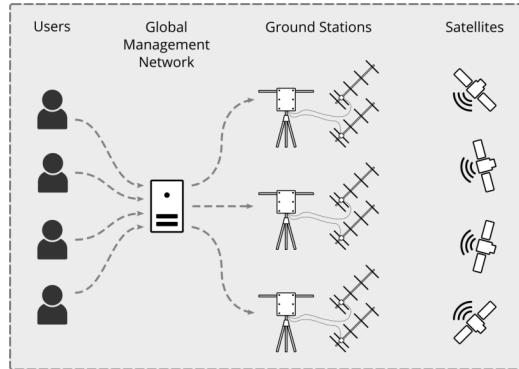
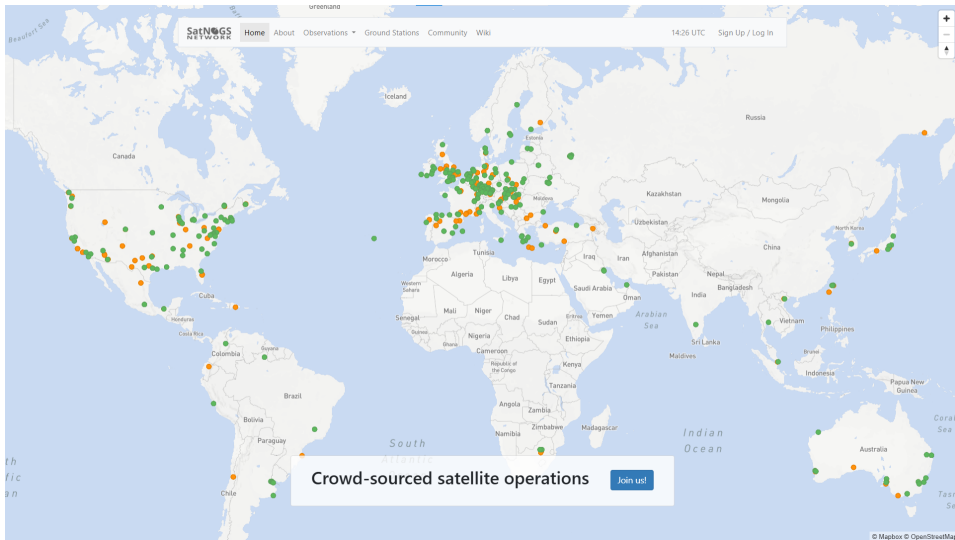


Figure: overall view of satnogs (Satellite Network Open Ground Stations) concept



satnogs network



bad results

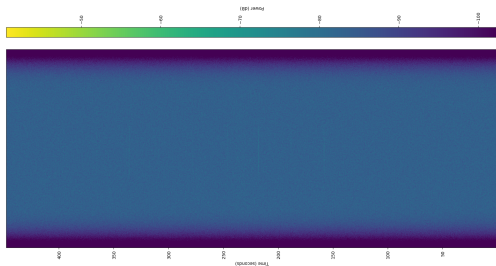


Figure: bad result – no signal

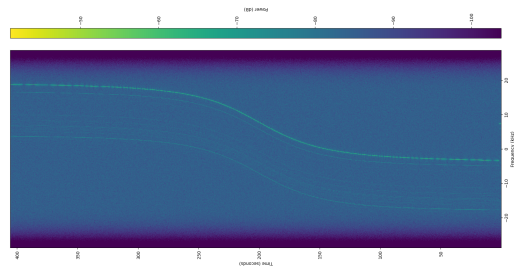


Figure: also a bad result – tangent-like signal means terrestrial noise



G A R A .
G E . O F
C O M . .
. P L E X
. . I T Y

project co-funded by Garage of Complexity



bibliography



thx for attention



<https://github.com/falcotinnunculus/przezrocza/blob/wladca/sempowisko2025/sempo.pdf>

